

FR1FR2TTUG

FR1_FR2_Test_Tool DPD and Closed-loop QEC User Guide

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User guide
CONFIDENTIAL

Document information

Information	Content
Keywords	FR1FR2TTUG, FR1, FR2, FR1_FR2_Test_Tool
Abstract	The document describes how to perform QEC calibration and RX IQ correction. It covers the detailed steps for closed-loop DPD training from time-domain and frequency-domain waveform sources.



1 Introduction

FR1_FR2_Test_Tool is a software tool running on the LA12xx device VSPA cores. It can:

- Generate 5G signals or single tone on LSDCS for sub 6 GHz FR1 to test RF card performance
- Generate 5G signals or single tone on HSDCS for mmwave FR2 to test RF card performance

The document describes how to perform QEC calibration and RX IQ correction. It covers the detailed steps for closed-loop DPD training from time-domain and frequency-domain waveform sources.

2 Environment setup

The following is a list of hardware and software components that are used for the setup.

- Baseband platform: LA1224-RDB-B-LS
- RF platform: Diora-3QB2 (N78) Diora-3QC (N79)
- BSP: LA1224-RDB BSP 2.4
- RF driver: Diora driver 2.1.1
<https://metanoia-comm.atlassian.net/servicedesk/customer/portal/3>
- FR1_FR2_TOOL test tool script version: 4.0
- FR1_FR2_TOOL VSPA image version: 4.0.5
- Python: 3.6

[Figure 1](#) depicts the block diagram of the setup used.

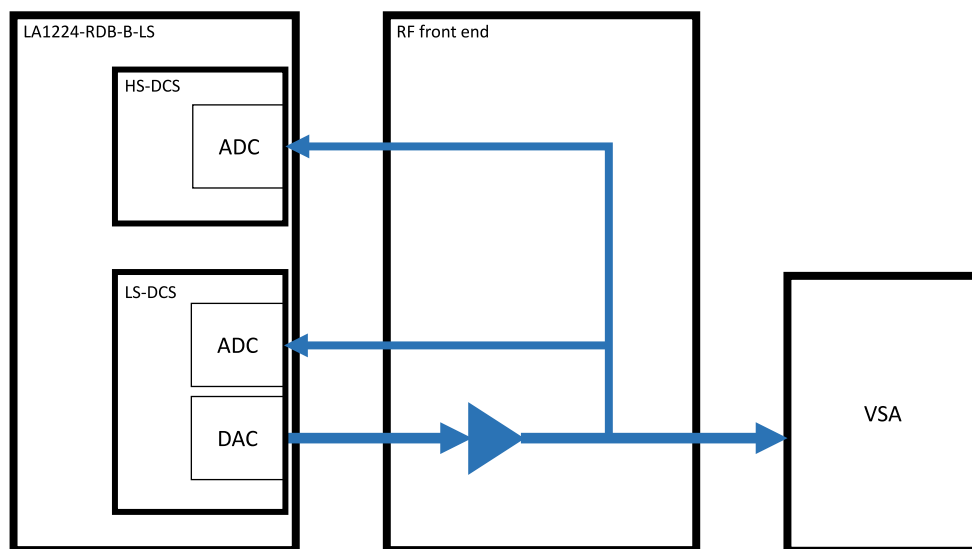


Figure 1. Block diagram

[Figure 2](#) shows the LA1224-RDB-B-LS board and Diora RF cards used on the board.

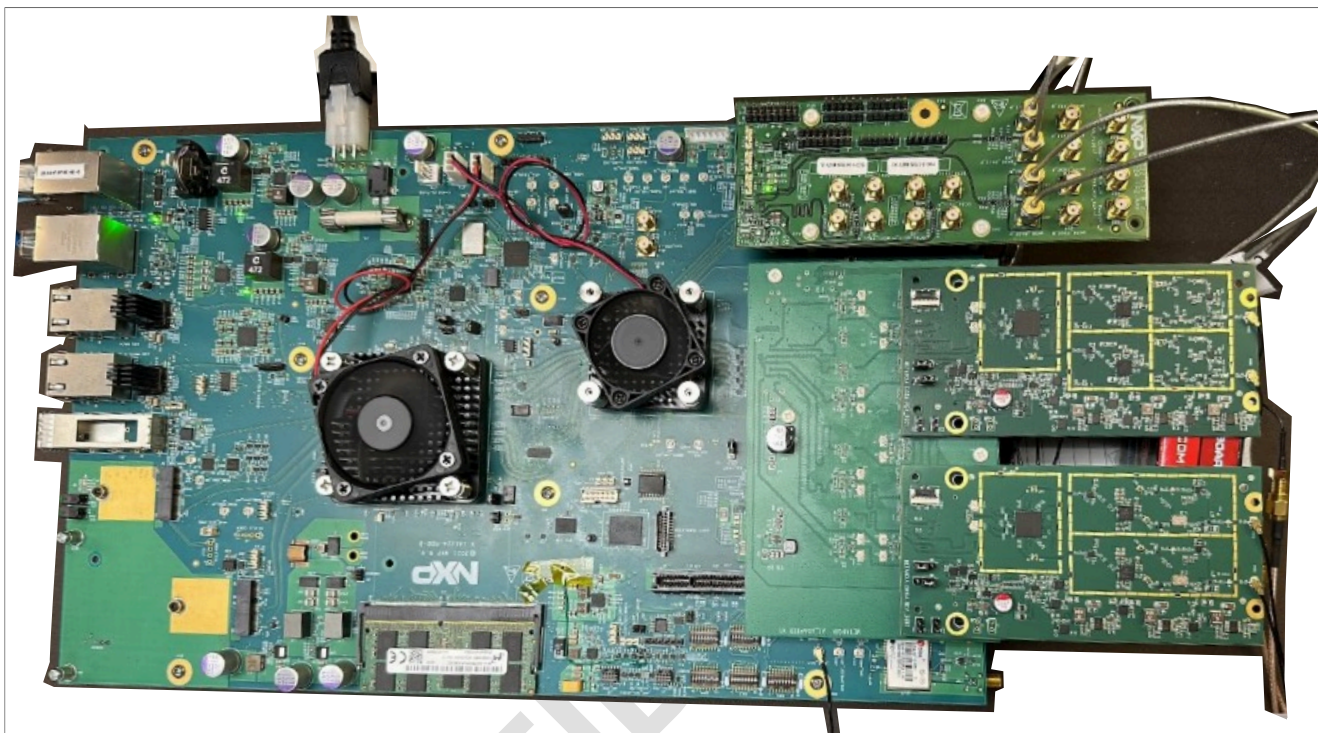


Figure 2. LA1224-RDB-B-LS and Diora RF cards

3 VSPA image prerequisites

The following list describes the prerequisites for the VSPA image.

- QEC calibration:
No prerequisites required
- RX IQ correction:
No prerequisites required
- DPD training: Time-domain waveform source
ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
 - LS training: Observation on LS ADC to capture waveform at 245.76 Msps
 - HS training: Observation on HS ADC to capture waveform at 983.04 Msps
- DPD training: Frequency-domain waveform source
ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD
 - LS training: Observation on LS ADC to capture waveform at 245.76 Msps
ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
 - HS training: Observation on HS ADC to capture waveform at 983.04 Msps

Note: The performance of DPD is affected by the signal quality, LO leakage, training sampling rate, and PAPR. For details about further RF performance tuning, contact RFIC vendors.

4 QEC calibration

The following is an example of running TRX DC/LO calibration and generating QEC coefficients by running the NXP QEC calibration tool.

Note: The TRX QEC calibration only works in RF loopback mode.

4.1 Enable FR1_FR2_Test_Tool

Run the following commands to enable FR1_FR2_Test_Tool.

```
cd fr1_fr2_test_tool/
./boot_vspa.sh ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD/
```

4.2 Initiate channel

Run the following command to initiate the channel.

```
** ./channels_start.sh <dcsid> **
./channels_start.sh dcs2 *** Calibration on DCS2 only ***

OR

./channels_start.sh *** Calibration on all channels ***
```

4.3 Check antenna mapping

Run the following commands to check antenna mapping for TX and RX.

```
Ant Mapping TX:
ant_id dcsid dcs status
0 0 LSDCS0-0 RUNNING
1 1 LSDCS0-1 RUNNING
2 2 LSDCS1-0 RUNNING
3 3 LSDCS1-1 RUNNING

Ant Mapping RX:
ant_id dcsid dcs status
0 0 LSDCS0-0 RUNNING
1 1 LSDCS0-1 RUNNING
2 2 LSDCS1-0 RUNNING
3 3 LSDCS1-1 RUNNING
```

4.4 Initiate RF driver RFLOOP mode

The CL QEC calibration requires RF TX LO and RX LO to work independently on different frequencies for application to identify the LO leakage power of TX and RX. In this case for FR1, a frequency offset of 45 MHz has been selected.

```
./<path>/diora_rfloop_3qb2_trx1.sh ** Refer to Section 12
```

RX LO: 3.5 GHz

TX LO: 3.545 GHz (Mkr1)

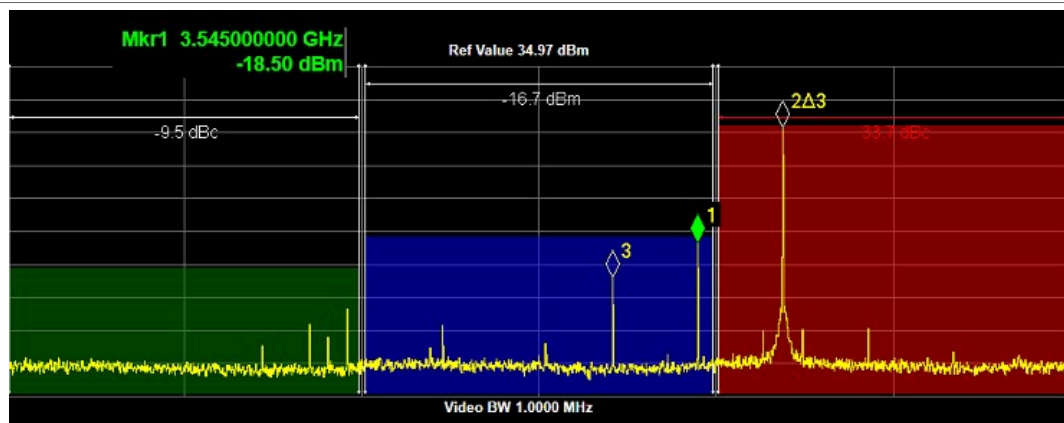


Figure 3. TX spectrum in RFLOOP mode

4.5 Initiate QEC calibration

Follow these steps to initiate QEC calibration.

1. Stop sending out 5GNR TM1.1 waveform
2. Transmit 24 MHz single tone waveform
3. Clean up the previous calibration data and perform QEC passthrough mode both on TX and RX
4. Calculate RX IQ imbalance and perform RX QEC: **rx_qec_cl.sh**
5. Calculate RX DC offset numbers and perform RX DC offset correction: **rx_dc_offset_cl.sh**
6. Calculate TX DC offset numbers and perform TX DC offset correction: **dc_offset_cl.sh**
7. Calculate TX IMRR and perform TX QEC: **tx_qec_cl.sh**
8. Stop sending single tone
9. Transmit 5GNR TM1.1 waveform

```
*** ./demo_qec_closetloop.sh <Tx_ant_id> <Rx_ant_id> ***
./demo_qec_closetloop.sh 0 0
```

RX: rx_qec_cl.sh result

```
root@frr1_fr2_test_tool# ./demo_qec_closetloop.sh 0 0 1
*****
* Close Loop QEC Calibration Version v2.0.0 *
*****

deleting previous *.cfg done
send_single_tone.sh 0 24000000 35
Sending single tone 24000000 Hz with 35% scale to antenna 0 , SPS=491520 Ksps
vspa_mbox send 4 1 0x0a0e02cc 0x0c800000
Antenna 0 is sending ONE time-domain single tone(s) now.

Passthrough QEC coefficients written to path:tx ant:0 !
Passthrough QEC coefficients written to path:rx ant:0 !

start QEC RX
Converted to 24 MHz

### Rx IQ Imbalance API ###
```

```
Test tone frequency (Tx): 24 MHz
Tx-Rx LO delta: 45 MHz
```

```
Observed tone frequency (Rx): 69 MHz
Rx capture length: 8192 samples
Rx FFT length: 4096 samples
```

```
### Rx IQ Imbalance extraction ###
```

```
FFT bin frequency: 69.0MHz
IMRR: 47.23 dB
```

```
Gain imbalance: 0.0014 dB
Phase imbalance: 0.4994 deg
```

```
Custom QEC coefficients written to path:rx ant:0 !
```

```
real    0m1.359s
user    0m0.738s
sys     0m0.652s
rx_qec_cl.sh done
```

RX: rx_dc_offset_cl.sh

```
start RX DC Offset Calibration
Detected DC offset: (0.032518) + 1j*(-0.014163)
Applying Rx DC correction:
    --dre -.032518
    --dim .014163
Custom QEC coefficients written to path:rx ant:0 !
rx_dc_offset_cl.sh done
```

TX: dc_offset_cl.sh

```
step      = .00048828125000000000;
dc_offset_values = -92.751085
dc_offset_best  = -76.736269
dc_i_tmp  = -.00585937500000000000; dc_q_tmp  = .00439453125000000000
dc_i_best = -.00585937500000000000; dc_q_best = .00439453125000000000

***** Computed parameters *****
> dc_offset_values = -92.751085 dB
> dc_q_obtn       = .00439453125000000000
> dc_i_obtn       = -.00585937500000000000

real    0m20.952s
user    0m11.423s
sys     0m15.871s
dc_offset_cl.sh done
```

TX LO leakage before and after (Mkr1)

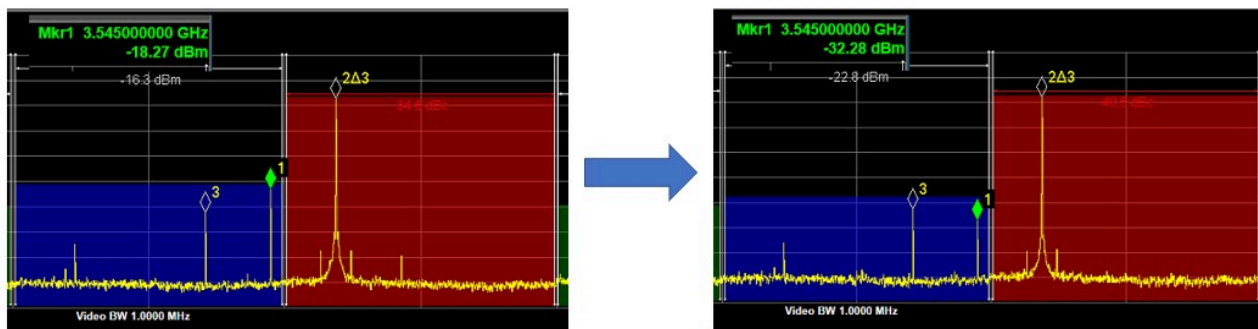


Figure 4. TX spectrum after IQ DC calibration

TX: tx_qec_cl.sh

```
***** Computed parameters *****
> IMRR_value           = 60.023 dB
> phase_obtn           = .62500000000000000000 degrees
> gain_obtn            = -.26562500000000000000 dB

real      0m17.883s
user      0m9.740s
sys       0m13.494s
tx_gec_cl.sh done
Stopping single tone to antenna 0
vspa_mbox send 4 1 0x0a0e21ff 0x00040000
Antenna 0 has stopped sending time-domain single tone.
```

Close Loop QEC Completed !

TX IMRR before and after (Mkr2)

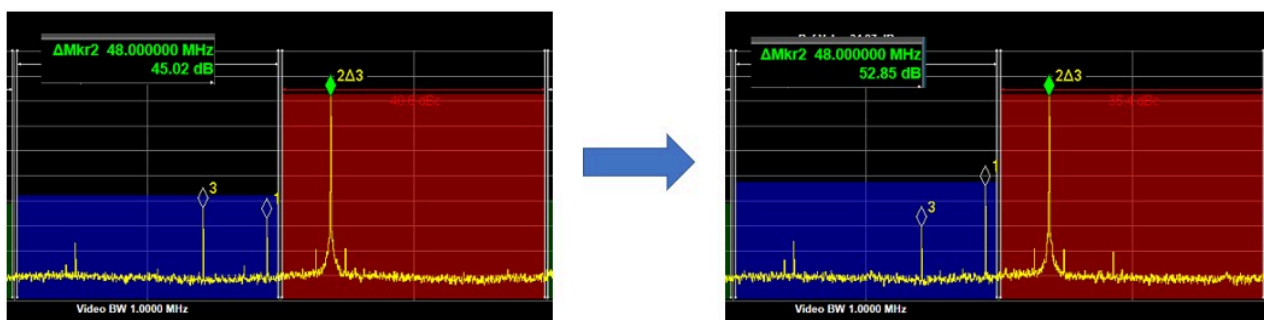


Figure 5. TX spectrum after IQ imbalance calibration

Calibration data of each antenna is stored as a .bin file in the fr1 fr2 tool/ folder.

- `gec_coeff_tx_<ant_id>.bin`
- `gec_coeff_rx_<ant_id>.bin`

These files are stored as a .cfg file in the fr1 fr2 tool/qec/ folder.

qec_tx_<ant_id>.cfg

```
declare -- iq_gain_imb_dB="0.16"
declare -- iq_phase_imb_deg="0.2"
declare -- FD_tap_cnt="0"
declare -- ssfilt_filename="0"
declare -- int_delay="0"
declare -- gain_re="1"
declare -- gain_im="0"
declare -- dc_re="-.001953125"
declare -- dc_im="-.0117187500"
```

qec_rx_<ant_id>.cfg

```
declare -- iq_gain_imb_dB="-0.201228221413"
declare -- iq_phase_imb_deg="1.16375211448"
declare -- FD_tap_cnt="0"
declare -- ssfilt_filename="0"
declare -- int_delay="0"
declare -- gain_re="1"
declare -- gain_im="0"
declare -- dc_re="0"
declare -- dc_im="0"
```

After QEC calibration, the QEC coefficients are applied on both TX and RX automatically.

You can update the QEC coefficient manually to check the EVM before and after.

- Before QEC check: apply QEC passthrough mode

```
./update_qec_coeff_tx.sh <ant_id> dis *** Tx ***
./update_qec_coeff_rx.sh <ant_id> dis *** Rx ***
```

- After QEC check: apply QEC coefficient from generated .bin

```
./update_qec_coeff_tx.sh <ant_id> qec_coeff_tx_<ant_id>.bin; *** Tx ***
./update_qec_coeff_rx.sh <ant_id> qec_coeff_rx_<ant_id>.bin; *** Rx ***
```

4.6 RX spectrum check

If the calibration is not working as expected, it is recommended to check the RX spectrum before or after calibration. You can do this by manually sending a 24 MHz single-tone waveform out of the DAC and then checking the FFT of the ADC dumped bin file.

```
./send_single_tone.sh <ant_id> 24000000 20
./dump_time_domain_rx.sh <ant_id>
python utils/plotdump.py -i rx_timedomain_20ms_245760ksps_dump_ant<ant_id>.bin -f 245760000
```

[Figure 6](#) shows the correct RX FFT plot.

- TX LO leakage: TX DC 45 MHz
- TX 24 MHz single tone power: 45 MHz + 24 MHz = 69 MHz
- TX 24 MHz single tone image power: 45 MHz - 24 MHz = 21 MHz

The above values should be identifiable without affected by distortion, frequency offset, or crosstalk.

Any of the above effects can downgrade the QEC performance.

For DC calibration, the DC power (TX LO leakage) is calculated and recorded.

For IQ imbalance calibration, the IMRR (TX 24 MHz single tone power - TX LO leakage) is calculated and recorded.

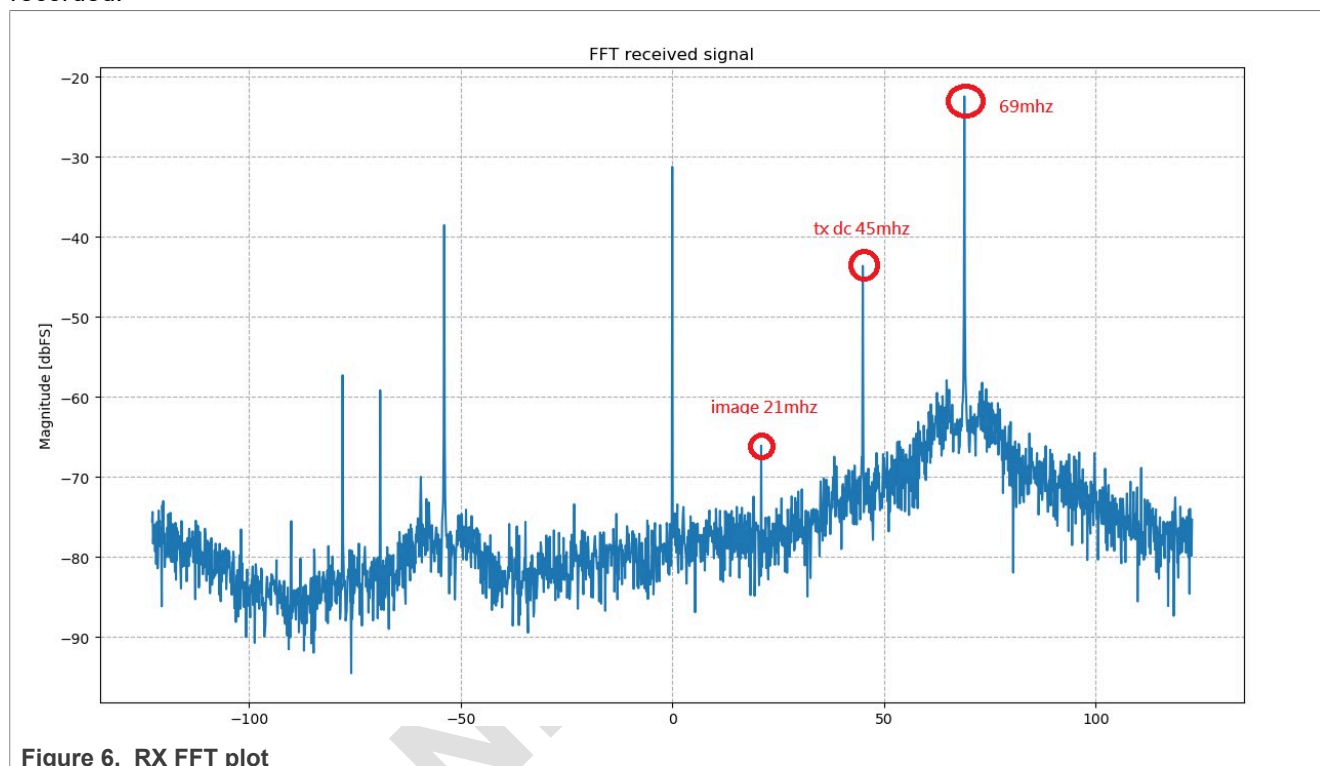


Figure 6. RX FFT plot

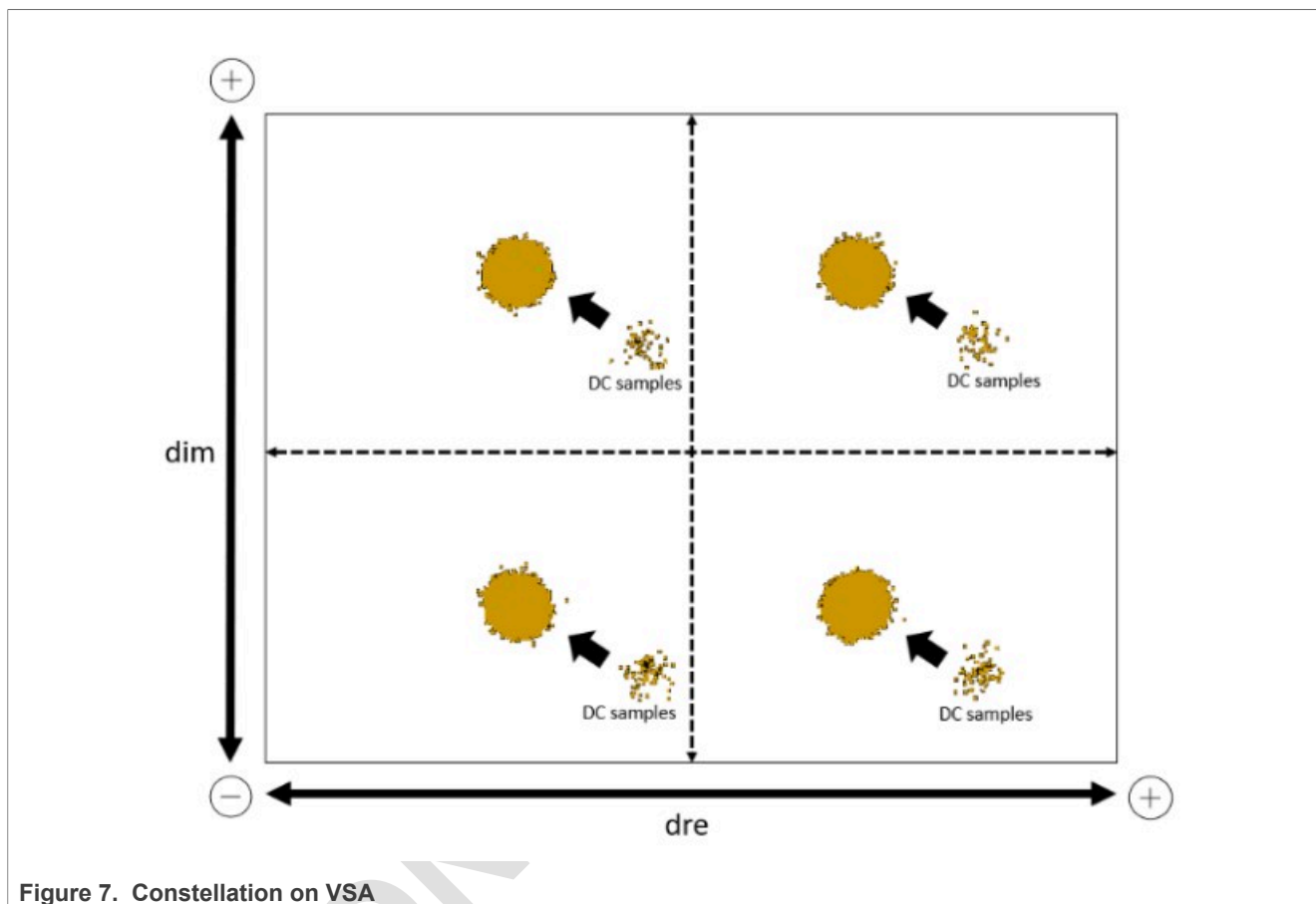
4.7 TX IQ fine tuning

Note: Be careful when adjusting the QEC tuning values, as improper inputs could potentially lead to boosted power and cause damage to the PA.

You can fine tune the DC samples by adjusting the DC offset and IQ imbalance values. However, this step is optional.

```
imb2qec_apply.sh -p tx/rx -a $tx/rx_ant_id --sg $gain_tmp --sp $phase_tmp --dre
$tx_corr_dc_re --dim $tx_corr_dc_im
```

You can monitor the constellation on VSA and try to align the constellation of DC samples with non-DC samples.



You can directly use the IQ imbalance reported by the VSA as a QEC input data.

The sign symbol should be changed for the QEC command.

For example:

- VSA report:
 - Gain imbalance: 0.23 dB
 - Quad error: -0.46 degree
- QEC command input:
 - --sg -0.23
 - --sp 0.46

Channel Power	20.25 dBm
Channel Power (Active)	20.25 dBm
EVM	5.40 %
EVM Peak	103.82 %
Frequency Error	1190.11 Hz
Symbol Clock Error	0.402 ppm
IQ Offset	-35.57 dB
Time Offset	-51 ns
Sync Correlation	99.9 %
Sync Source	PDSCH DMRS
Magnitude Error	3.68 %
Phase Error	0.04 rad
Gain Imbalance	0.23 dB
Quad Error	-0.46 degree
Timing Skew	-40.19 ps

Figure 8. Demodulation error summary

5 RX IQ correction

For some reason, if the RX is not well calibrated after QEC calibration, you can further fine-tune the RX performance by IQ correction.

The following is an example of how to measure the RX performance and run the RX IQ correction.

Note: The RX IQ correction only works in DPD Loopback mode.

5.1 Initiate RF driver DPD LOOP mode

Run the following command to initiate RF driver in DPD LOOP mode.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 11
```

5.2 RX measurement: Convert RX dump .bin to Keysight format

Run the following commands to convert the RX dump .bin file to Keysight VSA format.

```
./dump_time_domain_rx.sh 0
python3 utils/conv_bin2csv.py -i rx_timedomain_20ms_245760ksps_dump_ant0.bin -d
td -m fr1

Loading... rx_timedomain_20ms_245760ksps_dump_ant0.bin
>> TD Waveform sampling rate 245760000
resample from 245760000 to 122880000
1228800
>> Original TD waveform length in 20 slots = 1228800
>> Output .csv location =
rx_timedomain_20ms_245760ksps_dump_ant0_20231103013515.csv
```

On Keysight VSA, perform the following steps:

1. Recall >> Recording >> Recall From >> select .csv file

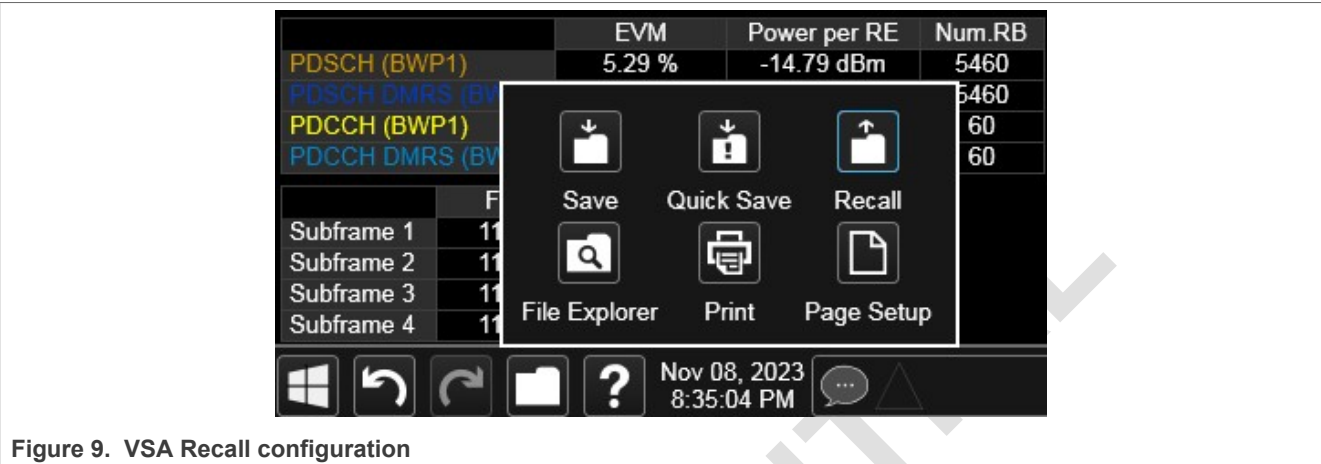


Figure 9. VSA Recall configuration

2. Meas Standard >> RB Alloc Preset >> DL NR-TM1.1 (P0) >> Apply Preset

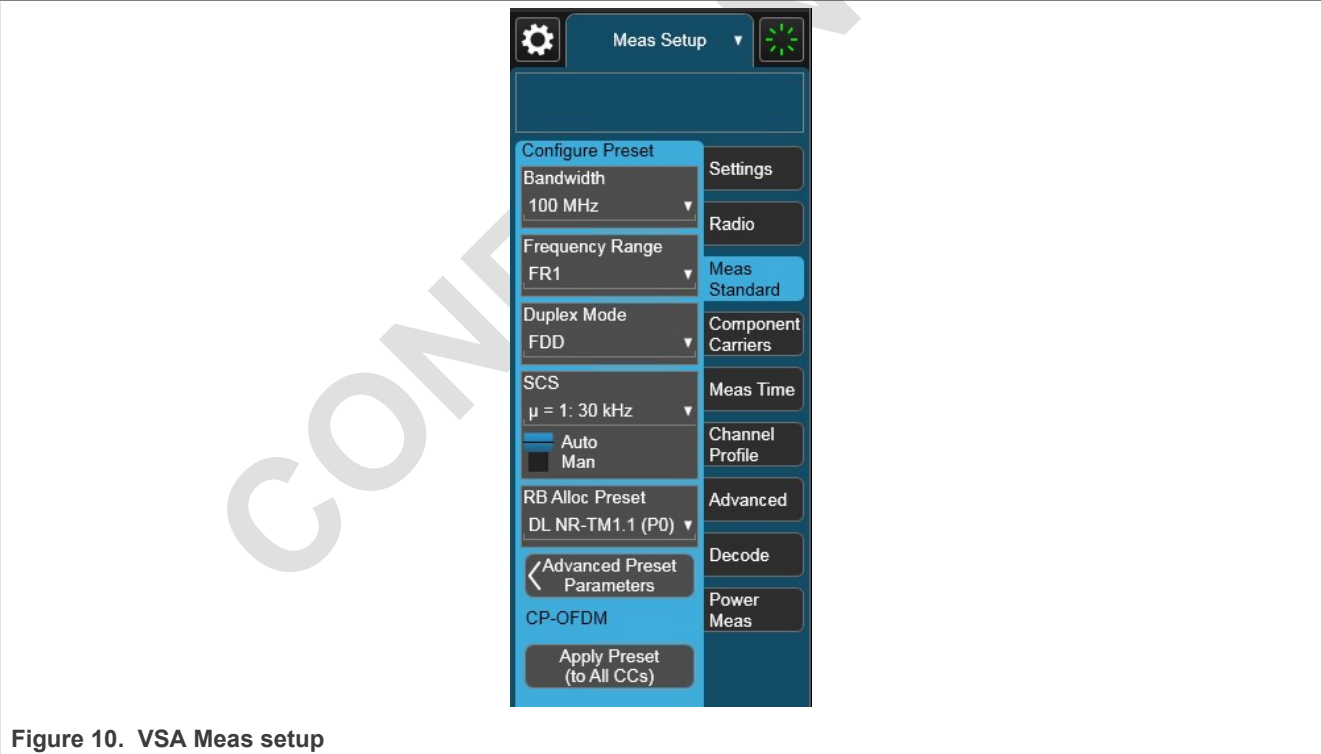


Figure 10. VSA Meas setup

In the below VSA snapshot, the RX EVM rms is 8.76 % and the EVM peak is approximately 280 %.

The OFDM samples affected by DC leakage cause distortion in the constellation and degradation in DPD performance.



Figure 11. RX EVM before calibration

5.3 IQ constellation correction

Run the following command to correct IQ constellation.

```
*** ./rx_iq_correction.sh <Tx_ant_id> <Rx_ant_id> ***

./rx_iq_correction.sh 0 0
Starting Rx IQ correction...
Custom QEC coefficients written to path:rx ant:0 !
incr_scale=4.512374878
1
IQ delta = 167294
Custom QEC coefficients written to path:rx ant:0 !
qec/imb2qec_apply.sh -p rx -a 0 --sg 0 --sp 0 --dre -.03313 --dim .01662
2
IQ delta = 15682
Custom QEC coefficients written to path:rx ant:0 !
qec/imb2qec_apply.sh -p rx -a 0 --sg 0 --sp 0 --dre -.03645 --dim .01763
3
IQ delta = 2056
Custom QEC coefficients written to path:rx ant:0 !
qec/imb2qec_apply.sh -p rx -a 0 --sg 0 --sp 0 --dre -.03667 --dim .01802
4
IQ delta = 1061
Custom QEC coefficients written to path:rx ant:0 !
qec/imb2qec_apply.sh -p rx -a 0 --sg 0 --sp 0 --dre -.03667 --dim .01780
5
IQ delta = 836
```

```
calibration done...found the best --dre -.03667 --dim .01780
```

5.4 Recheck calibrated RX

Repeat the step described in [Section 7.4](#) to play the RX dump on Keysight VSA.
The RX EVM peak is further improved from 280 % to 24 %.

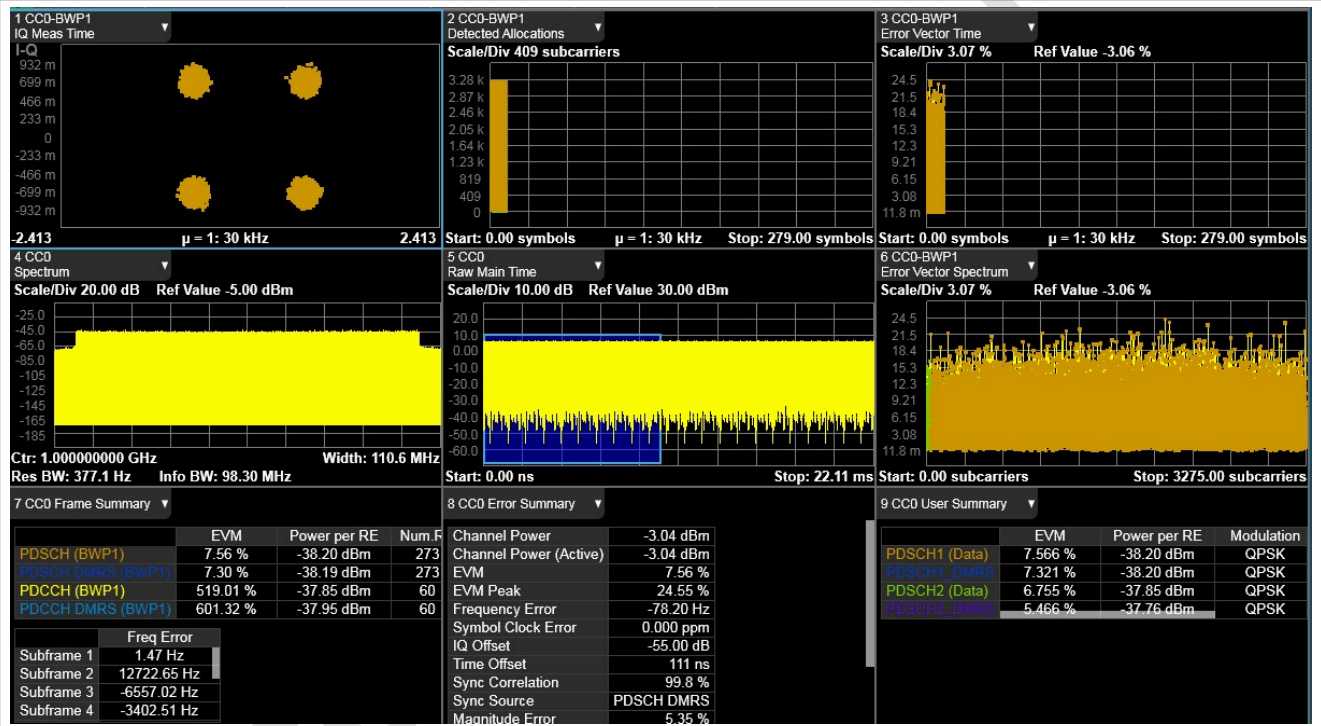


Figure 12. RX EVM after calibration

5.5 Different calibration combination

You can try different combinations on the calibration sequence to evaluate the performance.
The two examples are as follows.

Example 1 - RX IQ correction before DPD

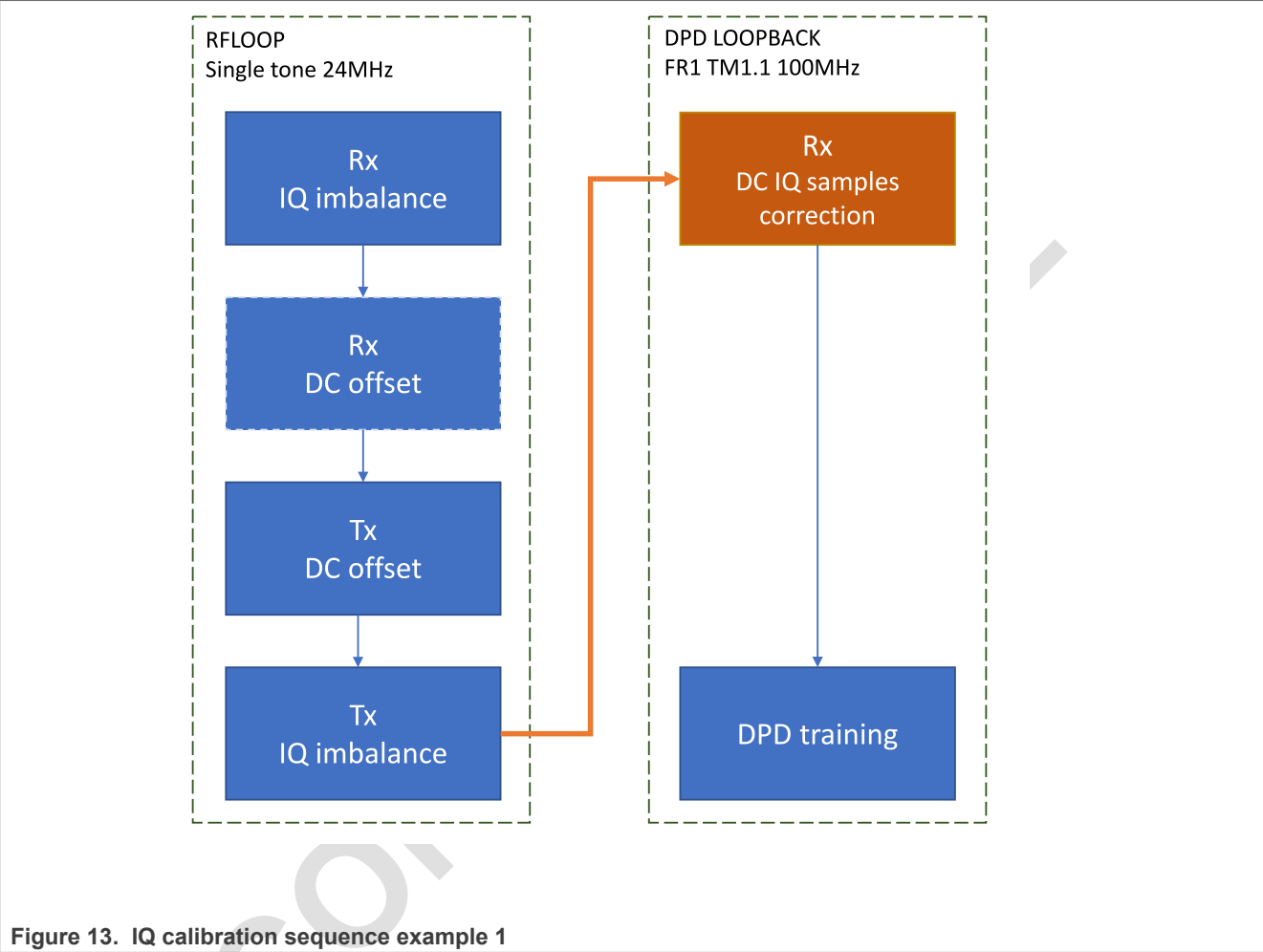


Figure 13. IQ calibration sequence example 1

Example 2 - RX IQ correction between each DPD training iterations

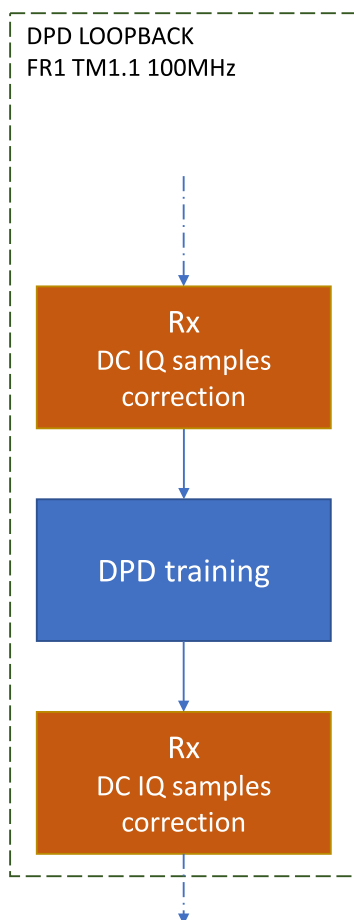


Figure 14. IQ calibration sequence example 2

6 Closed-loop DPD training from time-domain waveform source

6.1 Enable FR1_FR2_Test_Tool

Run the following command to enable FR1_FR2_Test_Tool.

```
./boot_vspa.sh
ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
```

6.2 Initiate channel

Run the following command to initiate channel.

```
** ./channels_start.sh <dcsid> **

./channels_start.sh dcs2 *** LS Training with observation path only on DCS2:
LS-ADC ***

OR
```



```
./channels_start.sh dcs2 dcs5 *** HS Training with observation path DCS4: HS-ADC ***
```

6.3 Check antenna mapping

Run the following commands to check antenna mapping for TX and RX.

```
Ant Mapping TX:
ant_id dcsid      dcs      status
0      2          LSDCS1-0  RUNNING

Ant Mapping RX:
ant_id dcsid      dcs      status
0      2          LSDCS1-0  RUNNING
```

6.4 Initiate RF driver DPD LOOP mode

Run the following command to initiate RF driver DPD LOOP mode.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 11
```

6.5 Apply QEC coefficient

Apply QEC TX and RX coefficients generated by QEC calibration.

```
./update_qec_coeff_tx.sh 0 qec_coeff_tx_<ant_id>.bin; *** Tx ***
./update_qec_coeff_rx.sh 0 qec_coeff_rx_<ant_id>.bin; *** Rx ***
```

6.6 Adjusting output signal power (if needed)

Run the following command to adjust output signal power.

```
./scale_percent.sh <ant_id> 140
```

6.7 Customize DPD model

You can choose one of the preset DPD models in config.dat. The following example shows how to use the customized DPD model R&D1 for DPD training. Note that only p can be changed in the r/p/q mapping.

```

# If a row is not needed, set p(max) to 0
use_custom_dpd_model=1 #valid only for T4x image

#This model is used by Advspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
dpd_model_para_custom=(
#r q p(max) example: 73 522 R&D1 R&D2 R&D2M
0 0 6 #max 6 6 4 6 5 5
1 0 3 #max 5 - 4 3 2 2
2 0 - #max 3 - - - - -
3 0 - #max 3 - - - - -
4 0 - #max 3 - - - - -
5 0 - #max 3 - - - - -
0 1 5 #max 5 - 4 5 4 4
1 1 5 #max 6 6 4 5 3 4
2 1 - #max 5 - - - - -
3 1 - #max 5 - - - - -
4 1 - #max 5 - - - - -
5 1 - #max 5 - - - - -
0 2 - #max 5 - - - 3 4
2 2 - #max 6 6 - - 3 3
0 3 - #max 5 - - - - -
3 3 - #max 5 - - - - -
0 4 - #max 5 - - - - -
4 4 - #max 5 - - - - -
0 5 - #max 5 - - - - -
5 5 - #max 5 - - - - -
)

```

Figure 15. DPD model configuration

6.8 Run DPD training

LS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 0 1 # 1 iteration
```

```
./dpd_training_close_loop.sh 0 0 3 # 3 iterations in a row
```

HS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 4 1 # 1 iteration
```

```
./dpd_training_close_loop.sh 0 4 3 # 3 iterations in a row
```

Check the training result and record the log.

```

root@fr1_fr2_test_tool# ./dpd_training_close_loop.sh 0 0 1
Ant 0 is set active.
DPD model: [ 0 0 6 ][ 1 0 3 ][ - - - ][ - - - ][ - - - ][ - - - ][ 0 1 5 ][ 1 1
5 ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ -
- ][ - - - ][ - - - ][ - - - ]

```

```

Dumping DPD output to file
tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
Using memory from 0xa06d5e8000 size 98304 as intermediate buffer for dumping...
Dumping feedback to file rx_timedomain_35840_245760ksps_dump_ant0.bin
Using memory from 0xa06d5f0000 size 65536 as intermediate buffer for dumping...

*****DPD CLOSE LOOP TRAINING RUN COUNTER = 1
*****
Ant 0 is set idle before training.
./dpd_training/dpdt ./dpd_training/dpd_spec.cfg 16384
tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
rx_timedomain_35840_245760ksps_dump_ant0.bin 1 0.00001
dpdt 1.7 (Aug 14 2023)
Copyright 2023 NXP
DPD model (GMP polyphase) structure (initialized from file ./dpd_training/
dpd_spec.cfg):
  Number of internal phases (R): 1
  GMP description: signal lag (r), envelope lag (q), poly order (p):

#  0  1  2  3
-----
r:  0  1  0  1
q:  0  0  1  1
p:  6  3  5  5

SRX measurements:
  avg power = 0.024804 = -16.0548 dBFS
  peak power = 0.230119 = -6.38047 dBFS
  PAPR = 9.6743 dB
Sync:.....
  54.6641 samples offset between ref and srx
  Level alignment gain: 1.50896
Diagonal loading: 1e-05
Training NMSE: -27.4315 dB
Training complete, coefficients written to dpd_coeff_vspa.flp
Ant 0 is set active after training.
TX signal shut down before applying DPD coeff.
Converting DPD coeff struct to MDPD required struct.
Updating DPD coeff from file on antenna 0 from address 0x23616c1000 to vspa
Current DPD model 19: [ 0 0 6 ] [ 1 0 3 ] [ - - - ] [ - - - ] [ - - - ] [ - - - ] [ 0
 1 5 ] [ 1 1 5 ] [ - - - ] [ - - - ] [ - - - ] [ - - - ] [ - - - ] [ - - - ]
[ - - - ] [ - - - ] [ - - - ] [ - - - ] [ - - - ]
Dumping DPD output to file
tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
Using memory from 0xa06d5e8000 size 98304 as intermediate buffer for dumping...
TX signal resumed.
Dumping feedback to file rx_timedomain_35840_245760ksps_dump_ant0.bin
Using memory from 0xa06d5f0000 size 65536 as intermediate buffer for dumping...

DPD Training done.

```

6.9 DPD model searching

FR1_FR2_Test_Tool provides an approach to search a DPD model depending on different board design or PA type. The results are recorded including DPD model r/q/p mapping, ACLR numbers, and estimated core

load, which helps to deal with the trade-off on different system design. The chosen DPD model can be applied manually in config.dat for further adjustment.

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration> srch
```

Example:

```
./dpd_training_close_loop.sh 0 0 1 srch # 1 iteration
```

The searching process takes some time, ranging from several minutes to even half an hour. The duration of the process depends on the number of polynomials and the desired iterations.

Figure 16. DPD auto search result

```
aclr=-38.07 aclrmpy100=-3807
Previous DPD model: [ 0 0 3 ][ 1 0 4 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 5 ][ 1 2 5 ][ 2 2 6 ]

END of DPD model searching ROUND 1, total searched num of models 49. Result saved in file ./dpd_training/dpd_model_search_report2023
09130722
Best DPD model ROUND 1: [ 0 0 3 ][ 1 0 4 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 5 ][ - - - ][ 2 2 6 ] ACLR=-38.42 Est_core_load
50%
Please choose next step:
1 Continue next ROUND of searching,
0 Stop searching and apply the best model and check ACLR on equipment.
You choice: 1
```

At the end of the searching process, you can choose to:

Press 1 to continue another searching.

Or

Press 0 to stop and train with the best model.

The searching process continues round 2 after 20 s if there is no input for the next step choice.

The printed ACLR number in the report is derived from the ADC dump .bin file. It could be lower than the VSA reported numbers due to the impact of RX gain/attenuation/filter. You can find the report in the folder dpd_training/.

```
dpd_model_searched_idx_122=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ - - - ][ - - - ][ 2 2 4 ] ACLR_122=-37.62 Est_core_load 41%
dpd_model_searched_idx_123=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 1 ][ - - - ][ 2 2 4 ] ACLR_123=-38.48 Est_core_load 44%
dpd_model_searched_idx_124=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 5 ] ACLR_124=-39.16 Est_core_load 46%
dpd_model_searched_idx_125=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 6 ] ACLR_125=-38.92 Est_core_load 47%
dpd_model_searched_idx_126=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ - - - ] ACLR_126=-37.9 Est_core_load 39%
dpd_model_searched_idx_127=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 1 ] ACLR_127=-38.88 Est_core_load 41%
dpd_model_searched_idx_128=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 2 ] ACLR_128=-38.99 Est_core_load 42%
dpd_model_searched_idx_129=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 3 ] ACLR_129=-38.9 Est_core_load 44%
dpd_model_searched_idx_130=[ 0 0 3 ][ 1 0 5 ][ 2 0 1 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_130=-38.87 Est_core_load 47%
dpd_model_searched_idx_131=[ 0 0 3 ][ 1 0 5 ][ 2 0 2 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_131=-38.8 Est_core_load 49%
dpd_model_searched_idx_132=[ 0 0 3 ][ 1 0 5 ][ 2 0 3 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_132=-38.94 Est_core_load 50%
dpd_model_searched_idx_133=[ 0 0 3 ][ 1 0 5 ][ 2 0 4 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_133=-39.01 Est_core_load 51%
dpd_model_searched_idx_134=[ 0 0 3 ][ 1 0 5 ][ 2 0 5 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_134=-38.62 Est_core_load 52%
dpd_model_searched_idx_135=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 1 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_135=-39.0 Est_core_load 47%
dpd_model_searched_idx_136=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 2 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_136=-38.9 Est_core_load 49%
dpd_model_searched_idx_137=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 3 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_137=-38.93 Est_core_load 50%
dpd_model_searched_idx_138=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 4 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_138=-38.9 Est_core_load 51%
dpd_model_searched_idx_139=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 5 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_139=-39.06 Est_core_load 52%
dpd_model_searched_idx_140=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 1 ][ 2 2 4 ] ACLR_140=-38.88 Est_core_load 47%
dpd_model_searched_idx_141=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 2 ][ 2 2 4 ] ACLR_141=-38.96 Est_core_load 49%
dpd_model_searched_idx_142=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 3 ][ 2 2 4 ] ACLR_142=-38.99 Est_core_load 50%
dpd_model_searched_idx_143=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 4 ][ 2 2 4 ] ACLR_143=-38.98 Est_core_load 51%
dpd_model_searched_idx_144=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 5 ][ 2 2 4 ] ACLR_144=-38.73 Est_core_load 52%
Best DPD model ROUND 1: [ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR=-39.25 Est_core_load 45%
root@fr1_fr2_test_tool# ~
```

6.10 PAPR measurement

The PAPR of TM1.1_100MHz_30kHz_FDD_timedomain_491520ksps.bin is 9.68 dB. Since the nonlinearity of PA causes gain compression in higher input power, the measured PAPR of SRx could be compressed.

You can monitor the CCDF curve on VSA and compare the PAPR between TX waveform and SRx waveform to make sure the linearity over the DPD training process.

```
python utils/calc_papr.py -i tx_timedomain_20ms_491520ksps_dump_ant0.bin

Input file is " tx_timedomain_20ms_491520ksps_dump_ant0.bin
papr before CFR: 8.93
papr before CFR: 9.51 db
```

Note:

- *Proper RX gain setting on RF is needed for correct PAPR observation.*
- *ADC clipping could cause the observed PAPR to be lower than the actual value.*

7 Closed-loop DPD training from frequency-domain waveform source

Note:

- *Customize DPD model is not supported in frequency domain mode.*
- *DPD model searching is not supported in frequency domain mode.*

7.1 Enable FR1_FR2_Test_Tool

Run the following command to enable FR1_FR2_Test_Tool.

```
*** LS Training with observation path on LS-ADC ***
./boot_vspa.sh ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD/

OR

*** HS Training with observation path on HS-ADC ***
./boot_vspa.sh
ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD/
```

7.2 Initiate channel

Run the following command to initiate channel.

```
** ./channels_start.sh <dcsid> **
./channels_start.sh *** LS Training with observation path on LS-ADC ***

OR

./channels_start.sh dcs2 *** LS Training with observation path only on DCS2:
LS-ADC ***

OR

./channels_start.sh dcs2 dcs4 *** HS Training with observation path on DCS4:
HS-ADC ***
```

7.3 Check Antenna mapping

Run the following commands to check antenna mapping for TX and RX.

```
Ant Mapping TX:
ant_id dcsid      dcs      status
0      0          LSDCS0-0  RUNNING
1      1          LSDCS0-1  RUNNING
2      2          LSDCS1-0  RUNNING
3      3          LSDCS1-1  RUNNING
```

```
Ant Mapping RX:
ant_id dcsid      dcs      status
0      0          LSDCS0-0  RUNNING
1      1          LSDCS0-1  RUNNING
2      2          LSDCS1-0  RUNNING
3      3          LSDCS1-1  RUNNING
```

7.4 Initiate RF driver DPD LOOP mode

Run the following command to initiate RF driver DPD LOOP mode.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 11
```

7.5 Apply QEC coefficient

Apply the QEC TX and RX coefficients generated by the QEC calibration.

```
./update_qec_coeff_tx.sh <ant_id> qec_coeff_tx_<ant_id>.bin; *** Tx ***
./update_qec_coeff_rx.sh <ant_id> qec_coeff_rx_<ant_id>.bin; *** Rx ***
```

7.6 Run DPD training

LS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 0 1 # 1 iteration
```

```
./dpd_training_close_loop.sh 0 0 3 # 3 iterations in a row
```

HS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 4 1 # 1 iteration
```

```
./dpd_training_close_loop.sh 0 4 3 # 3 iterations in a row
```

8 DPD open-loop training

The open-loop DPD training can be a useful tool for engineering development or in situations where there is no DPD loopback path on the RF side. It is recommended to use a high sampling rate of instrument configuration when conducting the open-loop DPD training.

Requirements:

- A 10 ms pulse trigger is needed to synchronize between LA1224-RDB-B and VSA.
- Connect J526 connector to VSA trigger input.

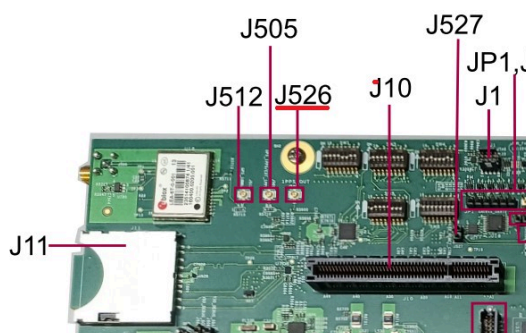


Figure 17. J526 location on LA1224-RDB-B

- Select the trigger source on VSA.

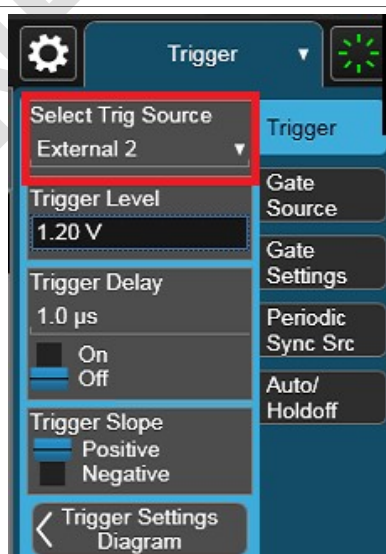


Figure 18. VSA trigger source selection

8.1 Enable FR1_FR2_Test_TOOL and RF

Enable FR1_FR2_Test_Tool and RF. For details, run the steps mentioned in [Section 6.1](#) to [Section 6.4](#).

8.2 Capture PA out data from VSA and record as .binx

Save >> Recording >> Save As >> All Supported Files >> .binx >> Save >> Recording_0000.binx

```
dd if=Recording_0000.binx of=Recording_0000.bin bs=512 count=1024
```

8.3 Generate DPD coefficient

Run the following command to generate DPD coefficient.

```
./dpd_training_open_loop.sh <ant_id> Recording_0000.bin <sample rate>  
./dpd_training_open_loop.sh 0 Recording_0000.bin 245760
```

9 Acronyms

[Table 1](#) lists the acronyms used in this document.

Table 1. Acronyms

Term	Definition
5G	Fifth-generation mobile communications
ADC	Analog-to-digital converter
BSP	Board support package
DAC	Digital-to-analog converter
DCS	Data conversion subsystem
DPD	Digital predistortion
eCPRI	Enhanced common public radio interface
GPS	Global positioning system
HS DCS	High-speed data conversion system
IMRR	Image rejection ratio
LO	FR1 center frequency
LS DCS	Low-speed data conversion system
NIC	Network interface controller
OFDM	Orthogonal frequency-division multiplexing
PA	Power amplifier
PCIe	Peripheral component interconnect express
QEC	Quadrature error correction
RF	Radio frequency
RX	Receiver
TDD	Time division duplex
TX	Transmitter
VSA	Vector signal analysis
VSG	Vector signal generation

Table 1. Acronyms...continued

Term	Definition
VSPA	Vector Signal processing accelerator

10 Related documentation

[Table 2](#) lists and explains the additional documents and resources that you can refer to for more information on LA12xxRDB and LA12xx-SDK. Some of the documents listed below may be available only under a non-disclosure agreement (NDA). To request access to these documents, contact your local NXP field applications engineer (FAE) or sales representative.

Table 2. Related documentation

Document	Description	Link / how to access
LA1224 Reference Design Board Getting Started Guide	Explains the LA1224RDB board settings and physical connections that are required to boot the board.	Contact NXP FAE / sales representative
LA1224 Reference Design Board Reference Manual	Provides a detailed description of the LA1224RDB board.	Contact NXP FAE / sales representative
LA1224 Reference Design Board (LA1224-RDB-B) Reference Manual	Provides a detailed description of the LA1224-RDB-B board.	Contact NXP FAE / sales representative
LA1224 Reference Design Board (LA1224-RDB-B) Getting Started Guide	Explains the LA1224-RDB-B board settings and physical connections that are required to boot the board.	Contact NXP FAE / sales representative
LA12xx Software Development Kit (LA12xx-SDK) User Guide	This document explains how to use the Layerscape Access LA12xx BSP for LA12xxRDB. It provides description of U-Boot, Linux, DPAA2 Ethernet interfaces, and FreeRTOS support on the LA12xxRDB and RF Subsystem.	Contact NXP FAE / sales representative

11 Appendix A: Diora 3QB2 RF DPDLOOP sequence example for LS DPD training

- diora_ls_dpd_3qb2_trx1.sh
- CF: 3.5 GHz
- TRX1 only

```
#!/bin/bash
echo 3 >/proc/sys/kernel/printk

function dio () {
    gul_refapp -c "$@" >/dev/null 2>&1
    echo run: $@
    sleep 0.5
}

dio "diora-close"
dio 'diora-open 0 0';
dio 'diora-version';
dio 'diora-setls 1';
dio 'diora-resetn 0';
dio 'diora-txrxsw 1';
dio 'diora-resetn 1';
```

```

dio 'diora-setpa 3 0';
dio 'diora-setdpd 3 0';
dio 'diora-setltna 3 0';
dio 'diora-txrxsw 1';
dio 'diora-txrxsw2 1';

echo Diora load
dio 'diora-load';
dio "diora-getsysstatus"

dio 'diora-calresistor 8';
dio 'diora-calregulator';
dio 'diora-settrxbw 11';
dio 'diora-settxbw 3';
dio 'diora-settrxp1l 1 7000000';
dio 'diora-setpath 5 2 0 1 11 3';
dio 'diora-setgain 4 1 0 1 19';
dio 'diora-setgain 1 1 0 1 7';
dio 'diora-setactive 1 1';
dio 'diora-txrxsw 0';
dio 'diora-txrxsw2 0';
dio 'diora-setpa 1 1';
dio 'diora-setltna 1 0';
dio 'diora-setdpd 1 1';
dio "diora-getsysstatus"

```

Note: For command details, contact the RF vendor.

12 Appendix A: Diora 3QB2 RF RFLOOP sequence example for LS QEC training

- diora_rfloop_3qb2_trx1.sh
- TX LO: 3.5 GHz
- RX LO: 3.455 GHz
- TRX1 only

```

#!/bin/bash

echo 3 >/proc/sys/kernel/printk

function dio () {
    gul_refapp -c "$@" >/dev/null 2>&1
    echo run: $@
    sleep 0.5
}

dio "diora-close"
dio "diora-open 0 0"
dio "diora-version"
dio "diora-setls 1"

dio "diora-resetn 0"
dio "diora-txrxsw 1"
dio "diora-resetn 1"
dio "diora-setpa 3 0"
dio "diora-setdpd 3 0"
dio "diora-setltna 3 0"

```

```

dio "diora-txrxsw 1"
dio "diora-txrxsw2 1"

echo Diora load
dio "diora-load"
dio "diora-getsysstatus"
dio "diora-calresistor 8"
dio "diora-calregulator"

dio "diora-setgain 4 1 0 1 16"
dio "diora-setgain 1 1 0 1 7"

dio "diora-settrxpll 1 7000000"

dio "diora-setrxbw 11"
dio "diora-settxbw 3"
dio "diora-setpath 5 2 0 0 11 3"

dio "diora-setcalpll 1 6910000"

dio "diora-setrfloop 1 2 1"
dio "diora-setdpd 3 0"
dio "diora-setlna 1 0"
dio "diora-setpa 1 1"
dio "diora-txrxsw 0"
dio "diora-txrxsw2 0"
dio "diora-getsysstatus"

```

Note: For command details, contact the RF vendor.

13 Appendix B: Diora 3QB2 n78 band LS DPD training result

```

DPD model
[ 0 0 5 ][ 1 0 5 ][ - - - ][ 3 0 1 ][ - - - ][ 5 0 3 ][ 0 1 5 ][ 1 1 0 ][ - -
- ][ 3 1 1 ][ - - - ][ - - - ]
[ - - - ][ - - - ][ 0 3 4 ][ 3 3 4 ][ - - - ][ - - - ][ - - - ][ 5 5 0 ]

```



Figure 19. ACP view of n78 after DPD training



Figure 20. EVM of n78 after DPD training

14 Appendix C: Diora 3QC n79 band LS DPD training result

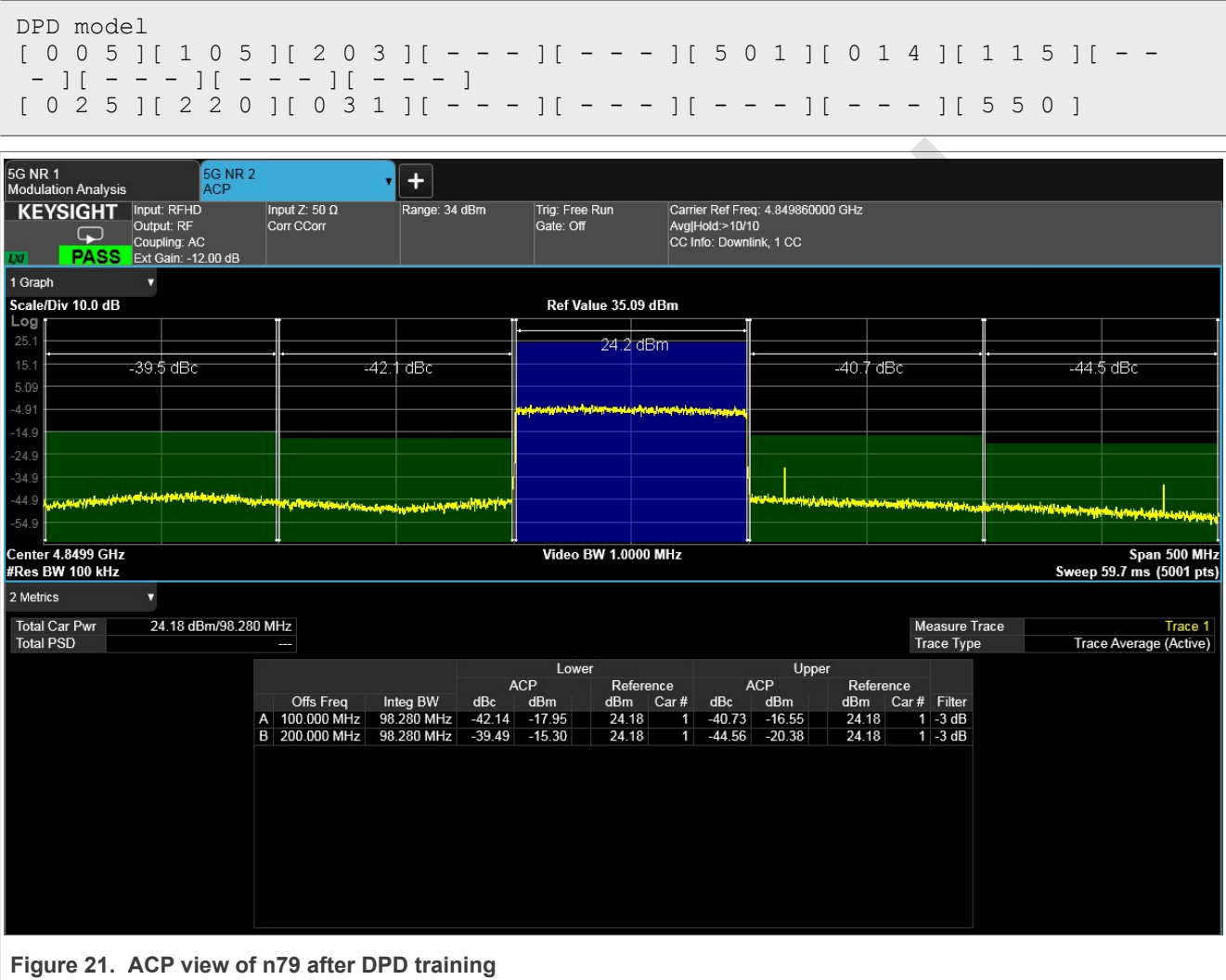


Figure 21. ACP view of n79 after DPD training

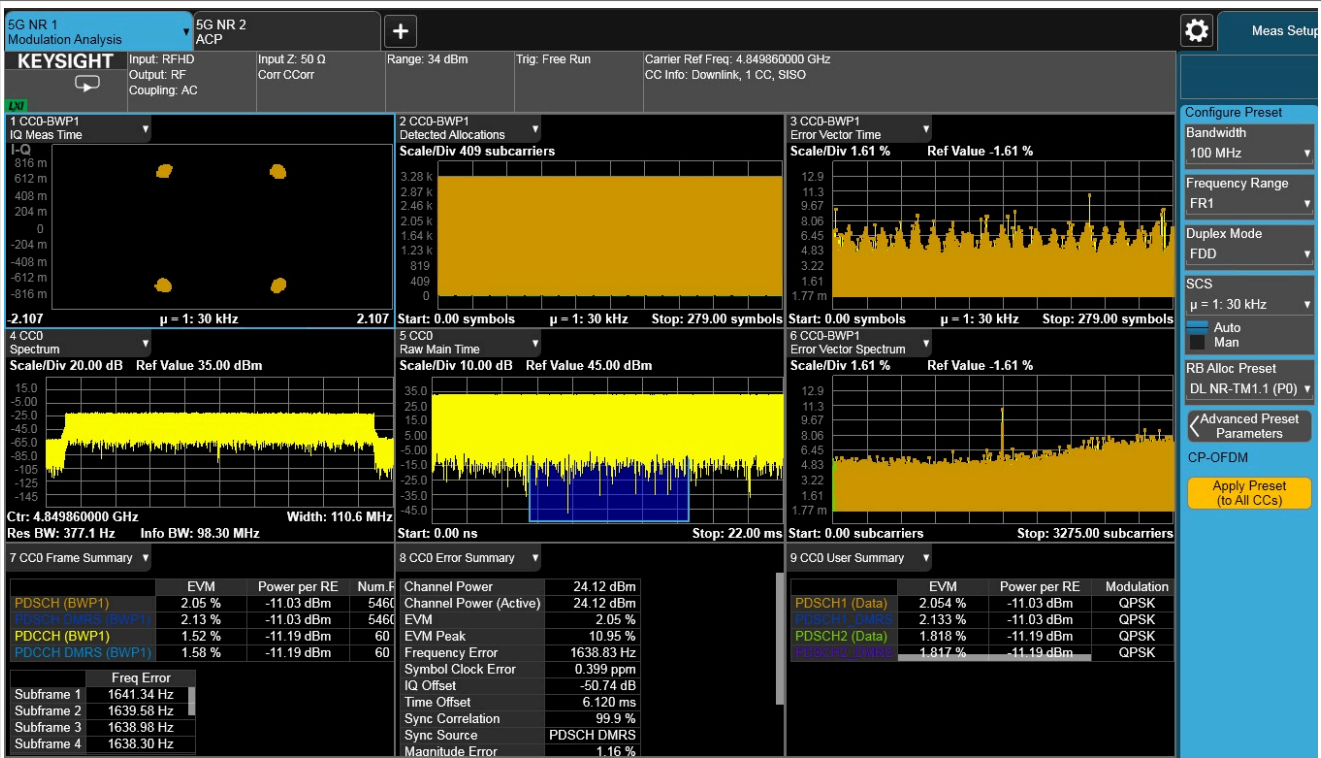


Figure 22. EVM of n79 after DPD training

15 Appendix D: RX overall performance after all calibration

- EVM rms: 1.29 %
- EVM peak: 12.62 %
- Gain imbalance: -0.03 dB
- Quad error: -0.19 degree

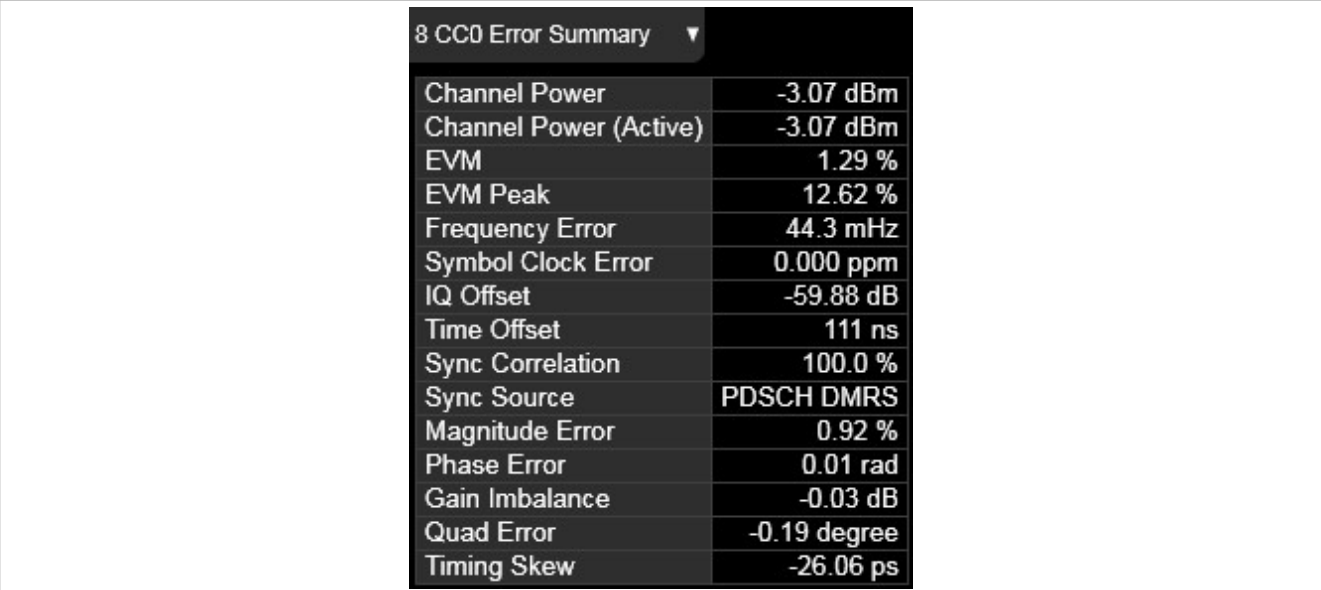


Figure 23. RX error summary

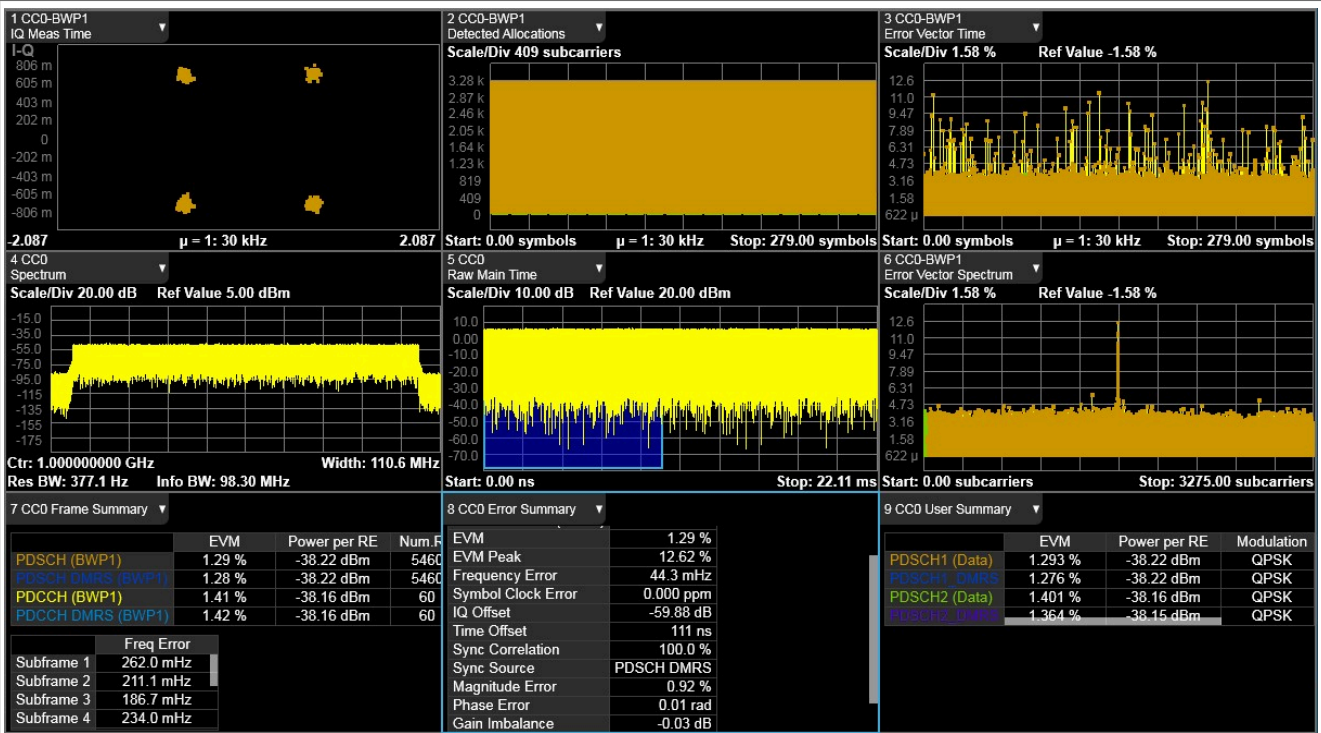


Figure 24. RX EVM

16 Revision history

Table 3 summarizes revisions to this document.

Table 3. Revision history

Revision number	Release date	Description
A	30 November 2023	Initial NDA release

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